SOLAR POWERED SMART STREET LIGHTING SYSTEM

A PROJECT REPORT

Submitted by

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in partial fulfillment for the award of the degree

of

BACHELOR OF ENGINEERING

IN

MECHANICAL ENGINEERING

KNOWLEDGE INSTITUTE OF TECHNOLOGY, SALEM

ANNA UNIVERSITY:: CHENNAI 600 025

JAN 2026

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BONAFIDE CERTIFICATE

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ABSTRACT

Solar-powered smart street lighting is an energy-efficient and sustainable solution designed to provide reliable outdoor illumination while reducing dependence on conventional grid electricity. This project focuses on the design and implementation of a Solar Powered Smart Street Lighting System, utilizing renewable solar energy, energy storage, and intelligent control mechanisms. The system operates by converting solar energy into electrical energy using photovoltaic panels, storing it in batteries, and efficiently powering LED street lights during nighttime operation. Smart features such as automatic light intensity control, motion detection, and dusk-to-dawn operation help minimize energy wastage and enhance system efficiency.The proposed system significantly reduces energy consumption by optimizing power usage and eliminating unnecessary lighting during low-traffic periods. High-efficiency LEDs are used to provide adequate illumination with minimal power requirements, while sensors and microcontroller-based controls ensure adaptive operation based on environmental conditions. The experimental model and performance analysis demonstrate improved energy efficiency, reduced operational costs, and enhanced reliability compared to conventional street lighting systems.In addition to improving public safety and visibility, the system contributes to environmental sustainability by lowering carbon emissions and promoting the use of renewable energy. The solar-powered smart street lighting system is particularly suitable for urban and rural areas, highways, and remote locations where grid power availability is limited or unreliable.

Keywords: Solar Energy, Smart Street Lighting, LED Lighting, Energy Efficiency, Sustainable Infrastructure.

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LIST OF SYMBOLS, ABBREVIATIONS AND NOMENCLATURE

SYMBOLS:

Symbol	Meaning
°C	Degrees Celsius
V	Voltage (Volts)
mA	Milliampere
kΩ	Kilo Ohms
MHz	Megahertz
GB	Gigabyte
W	Watts

ABBREVIATIONS:

Abbreviation	Full Form
PV	Photovoltaic
LED	Light Emitting Diode
LDR	Light Dependent Resistor
PIR	Passive Infrared Sensor
MCU	Microcontroller Unit
IoT	Internet of Things
PWM	Pulse Width Modulation
DC	Direct Current
AC	Alternating Current

NOMENCLATURE:

Term	Definition / Description
Solar Panel (PV)	Device that converts sunlight into electrical energy using photovoltaic cells
Battery Storage	System used to store electrical energy generated by the solar panel for night operation
LED Street Light	Energy-efficient lighting device used for street illumination
Charge Controller	Electronic unit that regulates voltage and current from the solar panel to the battery
Smart Lighting Control	Automatic adjustment of light operation based on environmental conditions
Light Dependent Resistor (LDR)	Sensor used to detect ambient light intensity for day/night control
Motion Sensor (PIR)	Sensor that detects movement to increase brightness when activity is detected
Energy Management	Efficient control and utilization of stored and generated energy
Autonomous Operation	System functioning independently without manual intervention
Solar Energy Utilization	Use of renewable solar power for system operation
Battery Protection	Methods to prevent overcharging and deep discharge of the battery
Dusk-to-Dawn Operation	Automatic switching ON at sunset and OFF at sunrise
Power Efficiency	Ability of the system to minimize energy losses during operation
Environmental Sustainability	Reduction of carbon emissions and environmental impact
Outdoor Operating Conditions	Environmental factors such as temperature, dust, and humidity affecting system performance

System Reliability	Ability of the system to operate continuously with minimal failure
Load Management	Control of power supplied to the LED based on system conditions
Energy Conservation	Reduction of unnecessary power consumption
Smart Monitoring (IoT)	Remote monitoring and data logging using internet connectivity (if applicable)
Thermal Management	Control of heat generated within system components
Digital Light Measurement	Detection of ambient light levels using sensors with digital output
Sensor-Based Monitoring System	System that collects environmental or operational data using sensors (LDR, PIR, temperature, etc.)
Real-Time Clock (RTC)	Module that maintains accurate date and time for scheduled operations
Time-Stamped Data Logging	Recording system parameters with corresponding date and time information
SD Card Data Storage	Storage of system performance or energy data on a removable memory card
High-Resolution Light Data	Light intensity or sensor data recorded with high precision
Interval-Based Logging	Recording system data at fixed time intervals (e.g., every minute or hour)
Prototype Development	Construction of a functional model to test and validate system performance
Adaptive Brightness Control	Automatic adjustment of LED brightness based on ambient light or motion detection
Energy Consumption Reduction	Minimization of electrical energy used by optimizing lighting operation

CHAPTER -1

1. INTRODUCTION

In Street lighting plays a vital role in modern infrastructure by ensuring road safety, enhancing public security, and improving visibility for vehicles and pedestrians during nighttime. Proper illumination of streets reduces the risk of accidents, deters criminal activities, and supports smooth transportation in both urban and rural areas. However, conventional street lighting systems are largely dependent on grid-based electrical power, resulting in high energy consumption, increased operational costs, and frequent maintenance requirements.

With the rapid growth of urbanization and expansion of road networks, the demand for street lighting has increased significantly. This rising demand places a heavy burden on electrical power generation systems and contributes to higher greenhouse gas emissions. In addition, frequent power outages and limited grid availability in rural and remote areas further reduce the reliability of traditional street lighting systems. These challenges highlight the urgent need for an alternative lighting solution that is energy-efficient, reliable, and environmentally sustainable.

Solar powered street lighting systems provide an effective solution by harnessing solar energy, a clean, renewable, and freely available resource. By converting sunlight into electrical energy through photovoltaic panels and storing it in rechargeable batteries, these systems can operate independently of grid power. The use of energy-efficient LED lamps further enhances system performance by providing high illumination with minimal power consumption and longer operational life.

The integration of smart control technologies significantly improves the efficiency of solar street lighting systems. Sensors such as Light Dependent Resistors (LDR) enable automatic switching of lights based on ambient light conditions, while motion sensors allow dynamic intensity control depending on vehicle or pedestrian movement. The objective of this project is to design and develop a Solar Powered Smart Street Lighting System that combines renewable energy generation with intelligent automation. The proposed system operates autonomously, adapts to environmental conditions, and ensures reliable illumination with minimal human intervention.

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1.1 METHODOLOGY

In this project, a Solar Powered Smart Street Lighting System is designed and evaluated using a systematic methodology. The main objective of this methodology is to provide efficient street illumination using renewable solar energy while minimizing electricity consumption and human intervention.

1. **Solar Resource Analysis**

   The availability of solar energy in the selected location is studied by analyzing sunlight intensity, duration of daylight hours, and seasonal variations. This analysis helps in selecting an appropriate solar panel capacity to ensure sufficient energy generation for reliable nighttime operation.

2. **System Design and Configuration**

   The overall system layout is designed by integrating major components such as the solar panel, rechargeable battery, LED street light, charge controller, sensors, and microcontroller. Proper configuration ensures efficient energy flow from solar energy collection to storage and utilization.

3. **Component Selection**

   High-efficiency components are selected to improve system performance and durability. Energy-efficient LED lamps are chosen for effective illumination with low power consumption. Suitable batteries are selected for reliable energy storage, and a microcontroller is used for intelligent control and automation.

4. **Sensor Integration and Smart Control**

   Sensors such as Light Dependent Resistors (LDR) and motion sensors are incorporated to enable smart operation. The LDR detects ambient light conditions to automatically switch the street light ON at night and OFF during daytime. Motion sensors control light intensity based on the presence of vehicles or pedestrians, reducing unnecessary power usage.

CHAPTER- 2

LITERATURE REVIEW

1. **R. Kumar et al. (2018)**

   The studied the design and performance of solar-based street lighting systems using high-efficiency LED lamps. Their research focused on energy consumption analysis and system reliability under varying environmental conditions. The study concluded that solar LED street lighting systems can reduce energy consumption by more than 60% compared to conventional grid-powered lighting systems, while also lowering maintenance requirements and improving operational lifespan.

2. **Santamouris (2020)**

   The emphasized the role of smart lighting systems in modern urban infrastructure and smart city development. The research highlighted the integration of motion sensors, automatic dimming, and adaptive lighting control to minimize energy wastage during low-traffic hours. The study demonstrated that motion-based lighting control significantly enhances energy efficiency and contributes to reduced carbon emissions in urban environments.

3. **Ali and Al-Alawi (2021)**

   The analyzed IoT-based smart street lighting systems with a focus on remote monitoring, fault detection, and system automation. Their work showed that integrating wireless communication technologies with solar-powered lighting enables real-time performance monitoring, faster maintenance response, and improved system reliability. The study concluded that smart automation reduces operational costs and enhances the scalability of street lighting infrastructure.

4. **Patel et al. (2022)**

   The investigated battery management and charge controller efficiency in solar street lighting applications. Their study highlighted the importance of proper energy storage and power regulation for ensuring uninterrupted lighting during extended cloudy conditions. The results indicated that efficient battery management systems significantly improve system durability and performance.

CHAPTER 3

COMPONENTS SPECIFICATION AND DESCRIPTION

3.1 COMPONENTS

The list of components used are,

S.NO	COMPONENTS	QUANTITY
1.	ESP32	1
2.	Rechargeable battery	1
3.	Battery holder	1
4.	Solar charge controller	1
5.	LED	1
6.	Solar panel	1
7.	LDR Sensor	1
8.	PIR Sensor	1
9.	Relay	1
10	Voltage Regulator	1
11	Switch	1
12	Jumper wire	1

Table 3.1 List of Components

3.2 COMPONENTS DESCRIPTION

3.2.1 ESP32 Board

The The ESP32 board is a low-power, high-performance microcontroller developed by Espressif Systems, widely used in Internet of Things (IoT) and embedded system applications. It integrates Wi-Fi and Bluetooth capabilities on a single chip, making it suitable for smart home and automation projects.

The ESP32 operates with a dual-core 32-bit processor and supports a clock frequency of up to 240 MHz. It provides multiple GPIO pins that can be configured for digital input/output, analog-to-digital conversion (ADC), pulse-width modulation (PWM), UART, SPI, and I²C communication. These features allow the ESP32 to interface easily with various sensors, actuators, and external modules.

One of the key advantages of the ESP32 board is its low power consumption, which includes several sleep modes such as deep sleep and light sleep. This makes it ideal for energy-efficient systems. The board also supports programming through the Arduino IDE, MicroPython, and ESP-IDF, offering flexibility for developers.

In this project, the ESP32 board acts as the main control unit. It is responsible for collecting data from connected sensors, processing the information, and controlling output devices according to the system requirements. Its built-in wireless communication enables remote monitoring and control, enhancing the overall functionality of the system.

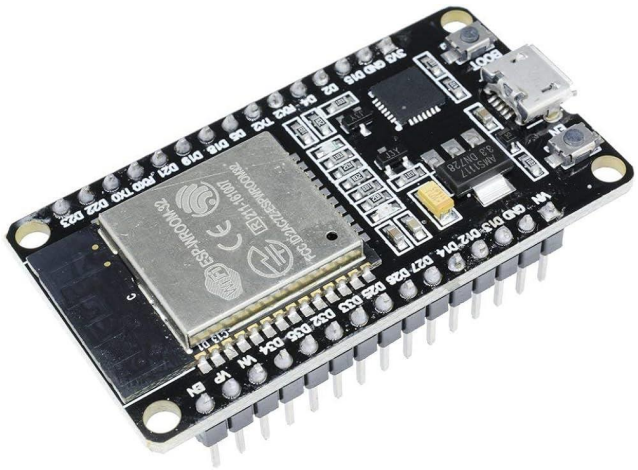


Figure 3.1 –ESP32 Board

3.1.1 RECHARGEABLE BATTERY

The rechargeable battery is a crucial energy storage component in the Solar Powered Smart Street Lighting System. It stores the electrical energy generated by the solar panel during daytime and supplies power to the LED street light and control circuitry during nighttime or low sunlight conditions.

The battery ensures continuous and reliable operation of the system even during cloudy days or extended periods without direct sunlight. Commonly used rechargeable batteries in solar lighting systems include lithium-ion or lead-acid batteries due to their high energy density, long cycle life, and reliability.

Proper charging and discharging of the battery are controlled by the solar charge controller to prevent overcharging, deep discharge, and voltage fluctuations. This protection increases battery lifespan and enhances overall system efficiency.

In this project, the rechargeable battery plays a vital role in maintaining uninterrupted illumination, improving energy utilization, and enabling the system to operate independently of grid power.



Figure 3.2 –RECHARGEABLE BATTERY

3.1.2 BATTERY HOLDER

The battery holder is an essential supporting component used to securely accommodate the rechargeable battery in the Solar Powered Smart Street Lighting System. It provides both mechanical stability and electrical connectivity, ensuring reliable power delivery to the system.

The holder is designed to maintain proper contact between the battery terminals and the circuit, reducing the risk of loose connections, voltage drops, or intermittent power supply. It also ensures correct polarity alignment, which helps prevent damage to electronic components due to reverse connection.

In addition to safety, the battery holder enhances ease of maintenance. The battery can be easily inserted or removed without disturbing other components, making replacement and inspection simple and time-efficient. The holder also provides basic insulation and protection against vibration, dust, and minor environmental disturbances.

In this project, the battery holder contributes to overall system reliability, safety, and long-term performance by ensuring secure battery placement and consistent electrical connectivity within the solar-powered lighting system.



Figure 3.3 – Battery holder

3.1.1 SOLAR CHARGE CONTROLLER

The solar charge controller is a key component in the Solar Powered Smart Street Lighting System, responsible for regulating the electrical power generated by the solar panel and safely charging the rechargeable battery. It acts as an interface between the solar panel, battery, and load, ensuring efficient and reliable energy management.

The charge controller prevents overcharging of the battery during peak sunlight conditions and protects against deep discharge during nighttime operation. By maintaining proper voltage and current levels, it significantly increases battery lifespan and improves system safety. The controller also protects the system from reverse current flow, which can occur when solar panel voltage drops at night.

In this project, the solar charge controller ensures optimal utilization of solar energy, stable battery operation, and continuous lighting performance. Its use enhances the overall efficiency, reliability, and durability of the solar-powered street lighting system.



Figure 3.4 – Controller

3.1.1 Jumper Wires

The Jumper wires are flexible, insulated wires used to make connections between components on a breadboard or between modules and microcontrollers like the ESP32. They come in male-to-male, male-to-female, and female-to-female types to suit different connection needs.

In this project, jumper wires are used to connect the ESP32 with sensors, the SD card module, and the RTC, enabling proper signal transmission and power supply. Their use ensures a neat, reliable, and easily modifiable circuit setup



Figure 3.5 – Jumper Wires

3.1.1 LED LIGHT

The 5 W LED light is used as the illumination source in the Solar Powered Smart Street Lighting System. LEDs are preferred due to their high luminous efficiency, low power consumption, and long operational life compared to conventional lighting sources such as incandescent or sodium vapor lamps.

The 5 W LED provides sufficient brightness for street lighting applications while consuming minimal electrical energy, making it highly suitable for solar-powered systems. It offers instant illumination, uniform light distribution, and minimal heat generation, which improves overall system efficiency.

In this project, the LED light is controlled by the ESP32 through a relay module, enabling automatic ON/OFF operation and smart intensity control based on ambient light conditions and motion detection. The use of a low-power LED ensures extended battery backup, reduced maintenance, and reliable nighttime illumination.



Figure 3.6 – Led light

3.1.2 LDR SENSOR

The Light Dependent Resistor (LDR) is a light-sensing component used to detect ambient light intensity in the Solar Powered Smart Street Lighting System. Its resistance varies according to the amount of light falling on its surface—low resistance in bright light and high resistance in darkness.

In this project, the LDR sensor is used to enable automatic operation of the street light. During daytime, when sufficient sunlight is present, the LDR signals the ESP32 to keep the LED light switched OFF. At night, when light intensity decreases, the increased resistance of the LDR triggers the controller to turn the LED light ON automatically.

The use of an LDR sensor eliminates the need for manual switching and reduces unnecessary power consumption. It improves system efficiency, reliability, and convenience by ensuring that the street light operates only when required.

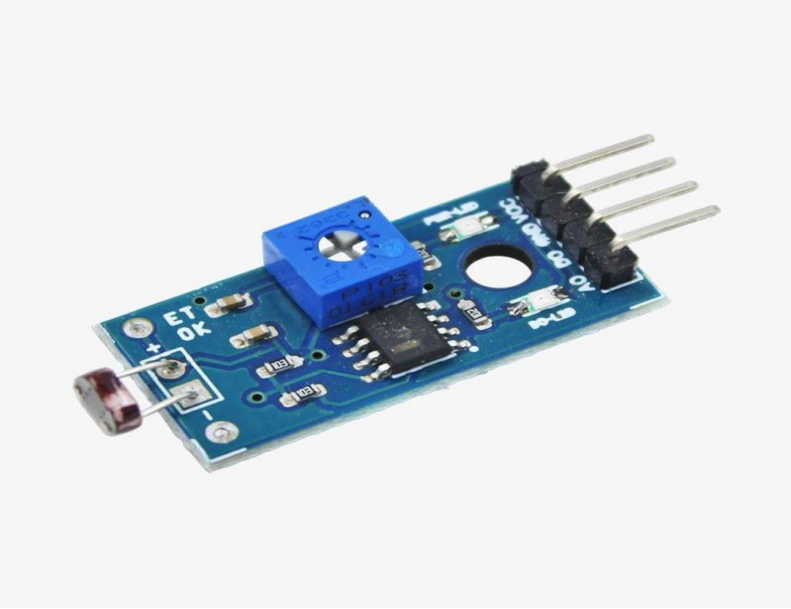


Figure 3.7 – LED SENSOR

3.1.1 PIR SENSOR

The Passive Infrared (PIR) sensor is used to detect motion in the Solar Powered Smart Street Lighting System. It works by sensing changes in infrared radiation emitted by moving objects such as vehicles or pedestrians within its detection range.

In this project, the PIR sensor enables smart and energy-efficient lighting operation. When motion is detected, the sensor sends a signal to the ESP32 microcontroller, which increases the light intensity or switches the LED light to full brightness. When no motion is detected for a certain period, the light returns to a low-power or dim mode, conserving battery energy.

The use of a PIR sensor significantly reduces unnecessary power consumption during low-traffic hours, extends battery backup duration, and enhances overall system efficiency. It also improves safety by providing adequate illumination only when required.



Figure 3.8 – PIR SENSOR

3.1.1 RELAY

The 5V relay is an electromechanical switching device used in the Solar Powered Smart Street Lighting System to control the LED light circuit. It allows a low-voltage control signal from the ESP32 microcontroller to safely switch a higher voltage or higher current load.

In this project, the relay acts as an interface between the control circuit and the LED lighting circuit. When the ESP32 sends a 5V control signal, the relay coil is energized, closing the contact and turning ON the LED light. When the signal is removed, the relay de-energizes and switches the light OFF.

The relay provides electrical isolation between the microcontroller and the power circuit, protecting sensitive electronic components from high current and voltage fluctuations. Its use ensures safe, reliable, and efficient operation of the smart street lighting system.

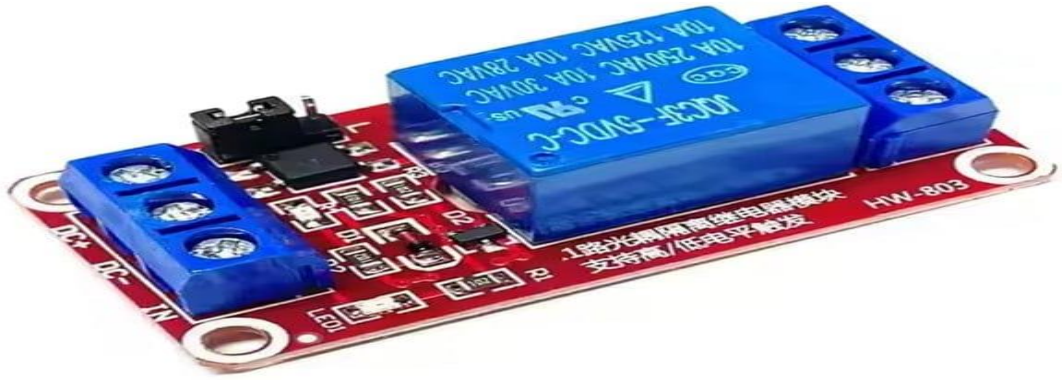


Figure 3.8 – PIR SENSOR

3.1.2 VOLTAGE REGULATOR

The voltage regulator is an essential component used to maintain a constant and stable output voltage in the Solar Powered Smart Street Lighting System. It ensures that sensitive electronic components such as the ESP32 microcontroller and sensors receive a regulated voltage supply, regardless of variations in battery voltage.

In this project, the voltage regulator steps down and stabilizes the battery voltage to the required operating level, typically 5 V or 3.3 V, suitable for the control circuit. This prevents damage to components caused by voltage fluctuations, overvoltage, or sudden drops in supply.

By providing a steady and reliable power source, the voltage regulator improves system stability, safety, and overall performance. Its use ensures smooth operation of the smart control unit and enhances the durability and reliability of the solar-powered street lighting system.

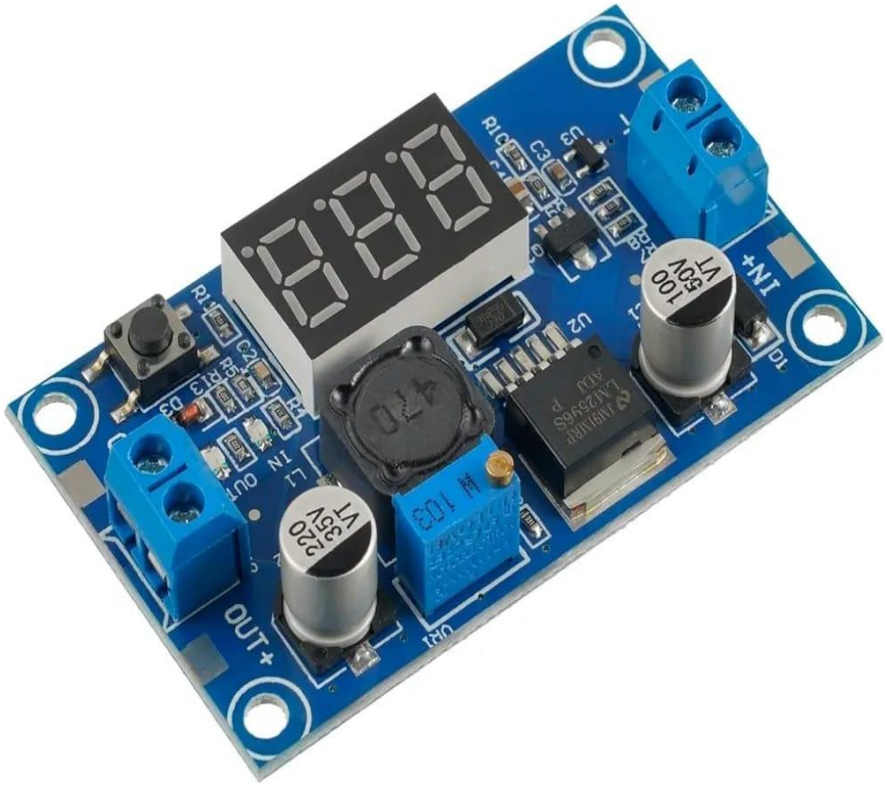


Figure 3.8 – VOLTAGE REGULATOR

3.1.1 SWITCH

The mini toggle switch is a manual control component used in the Solar Powered Smart Street Lighting System to turn the system ON or OFF as required. It provides a simple and reliable means of controlling the power supply during installation, testing, maintenance, or emergency situations.

In this project, the toggle switch is connected in series with the power supply to the control circuit. When the switch is turned ON, power is supplied to the ESP32 microcontroller and associated components, enabling normal system operation. When turned OFF, the power supply is disconnected, ensuring safety during maintenance and preventing unnecessary battery drain.

The use of a mini toggle switch enhances user control, system safety, and operational convenience, making it a practical addition to the solar-powered smart street lighting system.



Figure 3.8 – VOLTAGE REGULATOR

CHAPTER 4
CIRCUIT DESIGN AND CONNECTIONS

4.1 Introduction

The circuit design plays a vital role in the effective operation of the Solar Powered Smart Street Lighting System. The primary purpose of the circuit is to control the street light operation automatically using solar energy while ensuring efficient power management and intelligent lighting control.

The circuit is designed to be simple, reliable, and energy efficient, enabling uninterrupted operation with minimal maintenance. It integrates a solar panel, rechargeable battery, solar charge controller, microcontroller, sensors, relay module, and LED light to achieve autonomous street lighting functionality.

Each electronic component is selected based on compatibility, power efficiency, and ease of interfacing. The circuit ensures safe power distribution from the solar panel to the battery and from the battery to the control and lighting units. Smart sensors and the microcontroller work together to enable automatic ON/OFF operation, motion-based control, and efficient energy utilization, making the system suitable for long-term outdoor operation.

4.2 Circuit Diagram Description

The circuit diagram represents the interconnection of all major components used in the Solar Powered Smart Street Lighting System. The solar panel is connected to the solar charge controller, which regulates the charging of the rechargeable battery and protects it from overcharging and deep discharge. The battery serves as the main power source for the system during nighttime operation. A voltage regulator is used to provide a stable and regulated supply to the ESP32 microcontroller and sensor modules.

4.3 ESP32 Pin Configuration

The ESP32 microcontroller provides multiple GPIO pins supporting digital and analog communication protocols. Proper pin configuration is essential to ensure conflict-free operation and reliable control of the solar powered street lighting system.

S. No.	ESP32 Pin	Function	Connected Module
1	GPIO 34	Analog Input	Solar Panel Voltage Sensor
2	GPIO 35	Analog Input	Battery Voltage Sensor
3	GPIO 27	Digital Input	LDR Light Sensor
4	GPIO 26	Digital Output	LED Driver / Relay Module
5	GPIO 21 (SDA)	I²C Data Line	RTC Module (DS3231)
6	GPIO 22 (SCL)	I²C Clock Line	RTC Module (DS3231)
7	GPIO 5 (CS)	Chip Select	SD Card Module

4.4 Light Sensor (LDR) Connections

The Light Dependent Resistor (LDR) is used to detect ambient light levels for automatic ON/OFF operation of the street light. The LDR is connected in a voltage divider configuration to provide an among voltage proportional to light intensity.

- One terminal of the LDR is connected to 3.3 V
- The other terminal is connected to GPIO 27 through a fixed resistor
- The midpoint voltage is read by the ESP32 ADC

When ambient light decreases below a predefined threshold, the controller activates the street light automatically.

4.5 Solar Charge Controller Connections

The **solar charge controller** is used to regulate the voltage and current from the solar panel to the battery, ensuring safe and efficient charging. It protects the battery from overcharging, deep discharging, and short circuits, thereby improving battery life and system reliability.

Connections:

• Solar Panel (+) → PV+ terminal of Charge Controller

• Solar Panel (–) → PV– terminal of Charge Controller

• Battery (+) → Battery+ terminal of Charge Controller

• Battery (–) → Battery– terminal of Charge Controller

• Load (+) → Load+ terminal (connected to LED driver/ESP32 system)

• Load (–) → Load– terminal.

The charge controller continuously monitors the battery voltage and controls the charging process accordingly. During the daytime, it allows the solar panel to charge the battery. At night or during low sunlight conditions, it supplies stored energy from the battery to the street lighting system. This regulated power management ensures efficient energy utilization and enables reliable operation of the solar-powered smart street lighting system.

4.6 PIR Sensor Connections

The **PIR (Passive Infrared) sensor** is used to detect motion based on changes in infrared radiation emitted by moving objects such as pedestrians or vehicles. It is interfaced with the ESP32 to enable automatic brightness control and improve energy efficiency in the solar smart street lighting system.

Connections:

• VCC → 5 V (or 3.3 V, depending on module specification)

• GND → Common ground

• OUT → GPIO 27 of ESP32

The PIR sensor outputs a digital HIGH signal when motion is detected and LOW when no movement is present. Based on this signal, the system increases the LED brightness during activity and reduces it during idle periods.

4.7 LED Light Connections

The **LED light** is the primary illumination component of the solar powered smart street lighting system. It is controlled by the ESP32 through a driver or switching circuit to ensure efficient power utilization and reliable operation.

Connections:

• LED Positive (+) → Battery/Load positive through LED driver or MOSFET
• LED Negative (–) → MOSFET/Driver output (controlled by ESP32)
• Control Pin (Gate/Input) → GPIO 26 of ESP32 (PWM/Digital control)
• Power Supply → 12 V battery (through charge controller/load terminal)
• GND → Common groundThe LED operates based on control signals from the ESP32. It automatically turns ON during night conditions and OFF during daytime. Brightness can also be adjusted using PWM control or motion-based dimming to reduce energy consumption.

4.8 Jumper Wires Connections

Jumper wires are used to establish electrical connections between various components in the solar powered smart street lighting system. They provide a simple and flexible way to interconnect modules such as the ESP32, sensors, RTC, SD card module, and other peripheral devices during prototyping and testing.

CONNECTIONS:

• Male-to-Male wires → Used for connections on breadboards
• Male-to-Female wires → Used to connect ESP32 pins to module headers
• Female-to-Female wires → Used for module-to-module connections
• Power Lines → Used to distribute 3.3 V / 5 V supply and GND
• Signal Lines → Used for GPIO, I²C, SPI, and sensor signal connections

CHAPTER 5

WORKING PRINCIPLE

The Solar Powered Smart Street Lighting System operates as an integrated energy-efficient solution designed to provide automatic and reliable street illumination using renewable solar energy. The system combines solar power generation, energy storage, smart sensing, and embedded control to ensure effective lighting operation with minimal energy wastage.

The system collects solar energy during daytime through a photovoltaic panel and stores it in a rechargeable battery. During nighttime, the stored energy is used to power the LED street light. Intelligent control using sensors and a microcontroller enables automatic switching and adaptive lighting based on environmental conditions and movement detection.

The working principle is based on three major operational stages:

(1) Solar Energy Collection and Storage , (2) Automatic Night-Time Lighting Operation, (3) Smart Control and Energy Optimization

5.1 SOLAR ENERGY COLLECTION AND STORAGE

- **Solar Panel Orientation:** The solar panel is optimally oriented and tilted to receive maximum sunlight throughout the day, improving energy generation efficiency.
- **Solar Radiation Absorption:** Photovoltaic cells absorb solar radiation and convert it into electrical energy using the photovoltaic effect.
- **Energy Regulation:** The solar charge controller regulates the generated power to ensure safe and efficient charging of the rechargeable battery.
- **Battery Energy Storage:** Electrical energy produced during daytime is stored in the rechargeable battery for nighttime operation.
- **Overcharge Protection:** The charge controller prevents overcharging of the battery during peak sunlight hours.
- **Smart Control Unit:** A microcontroller-based system automatically controls ON/OFF operation, brightness levels, and energy usage based on time or environmental conditions.
- **LED Lighting Module:** Energy-efficient LED lamps provide bright illumination with low power consumption, increasing system lifespan and reducing maintenance costs

5.2 AUTOMATIC NIGHT-TIME LIGHTING OPERATION

Once sufficient solar energy is stored during daytime, the system automatically manages lighting during night hours:

1. Light Sensing Mechanism:

- Light Dependent Resistors (LDRs) detect ambient light levels and distinguish between day and night conditions.
- When daylight decreases below a preset threshold, the system automatically switches ON the street lights.
- At sunrise, the sensor triggers automatic switch-OFF, preventing unnecessary power consumption.

2. Smart Brightness Control:

- LED brightness is adjusted based on traffic or pedestrian movement using motion sensors.
- Lights operate at reduced intensity during low-activity hours to conserve energy indoor.
- Full brightness is restored instantly when motion is detected.

3. Energy-Efficient Operation:

- Stored solar energy is efficiently utilized throughout the night without grid dependency.
- Automatic control minimizes battery discharge and extends battery life.
- The system ensures reliable illumination while reducing operational and maintenance costs.

5.3 SMART CONTROL AND ENERGY OPTIMIZATION

The system intelligently manages energy generation, storage, and usage to ensure efficient and reliable operation:

- The control unit monitors solar power generation and battery status in real time.
- Energy is optimally distributed to lighting units to prevent overcharging or deep battery discharge.
- Adaptive control algorithms adjust light intensity based on time, traffic density, and ambient conditions.
- Overall power consumption is minimized, extending battery life and improving system efficiency.

5.4 SYSTEM OPERATION FLOW

Step No.	Operation Description
1	Sunlight falls on the solar panel, and electrical energy is generated through photovoltaic conversion.
2	The generated energy is regulated and stored in the rechargeable battery via the charge controller.
3	Ambient light levels are continuously monitored using a light sensor (LDR).
4	When daylight reduces below the threshold, the control unit automatically switches ON the street light.
5	LED brightness is adjusted based on time or motion detection to optimize power usage.
6	Stored solar energy is efficiently utilized throughout the night to provide continuous illumination.
7	At sunrise, the system switches OFF the light and resets to begin the next charging cycle.

5.5 WORKING LOGIC INTERPRETATION

The system's operation follows a continuous cyclic logic to ensure efficient energy usage and reliable lighting performance:

5.5.1 Plan:

Design the solar panel capacity, battery size, LED rating, and control strategy based on lighting requirements and environmental conditions.

5.5.2 Check:

Monitor solar energy generation, battery charge level, ambient light conditions, and system performance during operation.

5.5.3 Act:

Adjust lighting schedules, brightness levels, or control parameters to optimize energy efficiency and extend system lifespan.

5.6 SUMMARY

The Solar Powered Smart Street Lighting System operates through a self-sustaining and intelligent workflow integrating solar energy harvesting, energy storage, smart control, and efficient lighting. By combining renewable energy sources with automated control mechanisms:

High accuracy

5.6.1 Grid electricity usage is eliminated, resulting in significant energy savings and reduced operational costs.

5.6.2 Automatic lighting control and optimized energy utilization ensure reliable illumination with extended battery life.

The overall working principle demonstrates an eco-friendly, low-cost, and sustainable approach to public lighting, providing consistent illumination with minimal maintenance and zero carbon emissions.

CHAPTER 6

COST ESTIMATION

S.NO	COMPONENTS	QUANTITY	PRICE
1	ESP32	1	350
2	Rechargeable battery	1	200
3	Battery holder	1	40
4	Solar charge controller	1	100
5	LED	1	20
6	Solar panel	1	600
7	LDR Sensor	1	60
8	PIR Sensor	1	150
9	Relay	1	100
10	Voltage Regulator	1	120
11	Switch	1	30
12	Jumper wire	1	80
		TOTAL	1850

CHAPTER 7

SCOPE OF FUTURE WORK

- Advanced solar panel technologies such as bifacial or high-efficiency **PV modules** can be integrated to improve energy generation.
- Integration of **smart sensors, IoT connectivity, and adaptive control algorithms** can enhance real-time monitoring and system automation.
- Energy storage performance can be improved using **advanced battery technologies** like lithium-ion or supercapacitors.
- Intelligent **data logging and remote diagnostics** can enable better fault detection and maintenance planning.
- Large-scale deployment in urban and rural areas can significantly **reduce electricity costs** while promoting renewable energy adoption.

7.1 ADVANTAGES

- Operates entirely on **renewable solar energy with zero grid dependency.**
- Environmentally friendly and **reduces carbon emissions.**
- **Low operating and maintenance cost** compared to conventional street lighting systems.
- Automatic operation improves **reliability and energy efficiency.**
- Long lifespan due to efficient **LED lighting and optimized battery usage**.
- Suitable for **highways, urban roads, and remote locations.**

7.2 APPLICATIONS

- Urban and rural street lighting.
- Highways and road intersections .
- Smart city and smart village projects.
- Parks, parking areas, and campuses.
- Industrial zones and residential layouts.

CHAPTER 8
RESULTS AND ANALYSIS

8.1 Introduction

This chapter presents the experimental results obtained from the solar powered smart street lighting system implemented on the prototype model and provides a detailed analysis of its performance. The primary objective of the experiment is to evaluate the effectiveness of solar energy harvesting, automatic night-time lighting operation, and energy optimization. The performance data collected using the embedded monitoring system is analyzed to study solar power generation, battery charging and discharging behaviour, and lighting operation under different environmental conditions. Graphical and tabular representations are used to clearly illustrate the efficiency and reliability of the system.

8.2 System Data Collection

System data is collected continuously at regular intervals using the embedded sensing and control modules. Voltage and current sensors monitor the solar panel output and battery status, while a light sensor (LDR) detects ambient light conditions for automatic ON/OFF control.

All measured parameters are recorded with accurate time stamps using the RTC module and stored on the SD card. Continuous monitoring enables detailed analysis of daily energy generation patterns and night-time lighting performance.

8.3 Daytime and Night-Time Performance Comparison

The collected data shows a clear difference between daytime and night-time system behaviour. During daytime hours, solar panel output increases rapidly due to direct sunlight, allowing effective energy generation and battery charging.

The results indicate that sufficient energy is stored during the day, enabling reliable lighting operation throughout the night. This performance difference demonstrates the effectiveness of solar energy harvesting and storage mechanisms.

8.4 Performance During Peak Sunlight Hours

The performance of the solar powered street lighting system is most evident during peak sunlight hours, typically between 11:00 AM and 2:00 PM. During this period, solar irradiance reaches its maximum level.

Experimental results show that the battery charging rate is highest during peak hours, ensuring adequate energy availability for night-time lighting. This confirms the efficiency of the solar panel and charge controller.

8.5 Battery Stability Analysis

In addition to efficient charging, the system also provides stable battery performance during night-time operation. The battery voltage shows minimal fluctuations, indicating controlled energy discharge.

The reduced rate of voltage drop during operation demonstrates effective energy management, which improves battery life and ensures continuous illumination.

8.6 Graphical Representation of Results

Graphs plotted using the recorded data show clear trends in solar voltage, battery charge levels, and lighting operation. The solar output curve exhibits peaks during daylight hours, while the battery discharge curve remains steady at night.

The graphical comparison highlights the transition between charging and discharging cycles, which is a key indicator of reliable smart control.

8.7 Effectiveness of Individual Passive System Components

The The experimental results suggest that each system component contributes to overall performance:

- **Solar Panel:** Efficient conversion of sunlight into electrical energy
- **Battery Storage:** Reliable energy storage for night-time use
- **Charge Controller:** Prevents overcharging and deep discharge
- **LED Lamp:** Provides high illumination with low power consumption

8.8 Energy Saving Potential

The system eliminates the need for grid electricity by operating entirely on solar energy. This directly results in significant energy savings and reduced electricity costs.

Automatic control and optimized energy usage make the system an energy-efficient and environmentally friendly lighting solution.

8.9 Limitations of the Experiment

- The experiment is conducted using a prototype-scale setup
- System performance depends on weather and sunlight availability
- Long-term battery degradation is not evaluated

Despite these limitations, the results provide strong evidence of the effectiveness and feasibility of the solar powered smart street lighting system.

CHAPTER 9

APPENDICES

9.1 PROTOTYPE



FIGURE 9.4



FIGURE 9.5



FIGURE 9.6

CONCLUSION

Thus, the design and development of a Solar Powered Smart Street Lighting System has been successfully completed, fulfilling all the objectives stated at the beginning of the project with satisfactory results. In this project, conventional grid-powered street lighting systems were replaced by an energy-efficient and renewable solar-based lighting solution that utilizes photovoltaic energy, battery storage, and intelligent control mechanisms. The system effectively provides reliable illumination during night-time hours by harvesting solar energy during the day and utilizing it efficiently after sunset, thereby eliminating dependency on conventional electrical power sources. The developed model integrates key components including solar panels, rechargeable batteries, charge controllers, LED lighting units, light sensors, and a microcontroller-based control system. These components work together to ensure automatic operation, optimized energy usage, and consistent lighting performance under varying environmental conditions. The system demonstrated stable operational behaviour, efficient battery charging during daylight hours, and uninterrupted lighting throughout the night. This project proves that solar powered street lighting is a cost-effective, sustainable, and environmentally friendly alternative to conventional street lighting systems, especially in urban, rural, and remote areas. The successful implementation of this model highlights its strong potential for large-scale deployment in smart cities, highways, residential layouts, and public infrastructure projects. Hence, the developed solar powered smart street lighting system represents an important step toward energy conservation, reduced carbon emissions, and sustainable development in modern society.

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